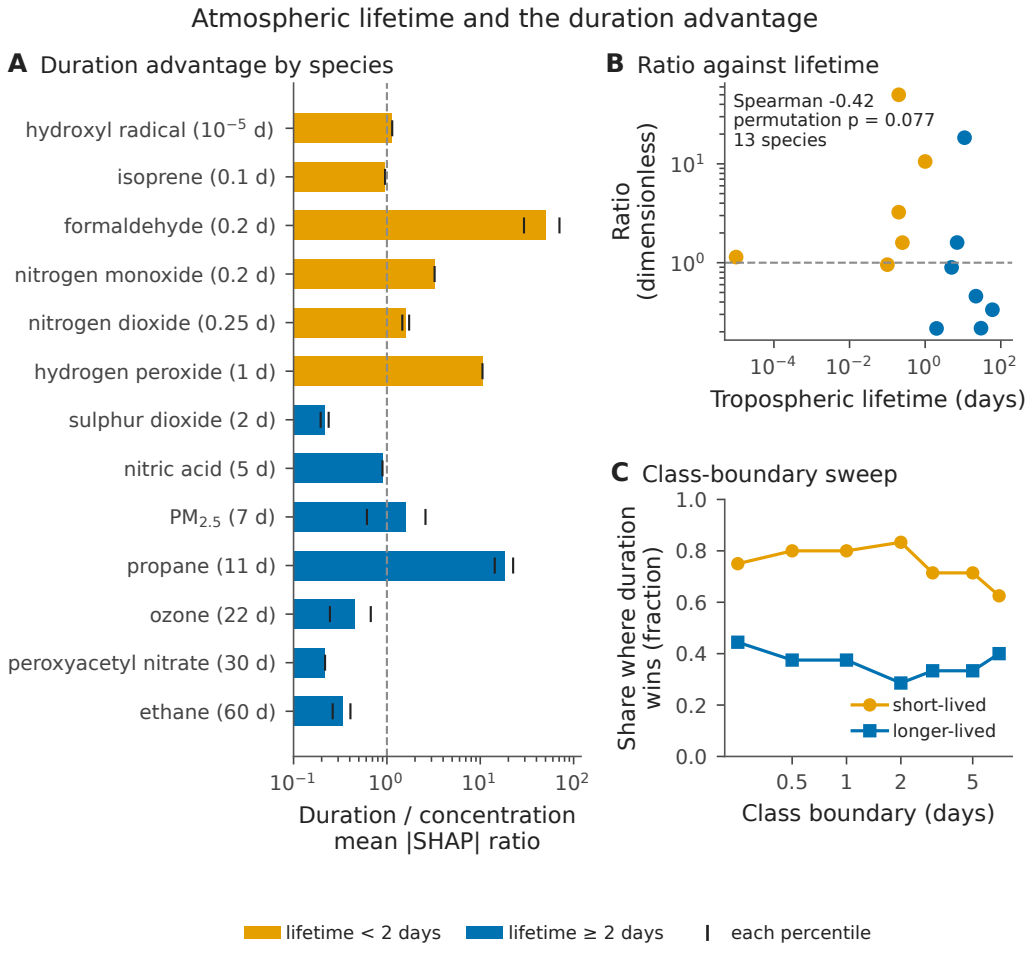
Atmospheric lifetime and the duration advantage
A Duration advantage by species
hydroxyl radical (10^{-5} d)
isoprene (0.1 d)
formaldehyde (0.2 d)
nitrogen monoxide (0.2 d)
nitrogen dioxide (0.25 d)
hydrogen peroxide (1 d)
sulphur dioxide (2 d)
nitric acid (5 d)
$PM_{2.5}$ (7 d)
propane (11 d)
ozone (22 d)
peroxyacetyl nitrate (30 d)
ethane (60 d)
10^{-1}
10^{0}
10^{1}
10^{2}
Duration / concentration
mean |SHAP| ratio
B Ratio against lifetime
Spearman -0.42
permutation p = 0.077
13 species
Ratio
(dimensionless)
10^{1}
10^{0}
10^{-4}
10^{-2}
10^{0}
10^{2}
Tropospheric lifetime (days)
C Class-boundary sweep
Share where duration
wins (fraction)
1.0
0.8
0.6
0.4
0.2
0.0
short-lived
longer-lived
0.5
1
2
5
Class boundary (days)
lifetime < 2 days
lifetime ≥ 2 days
each percentile